MATH 6610: Advanced Numerical Analysis

Homework 4

Due: Friday, December 4, 2026

Instructions. Show all work. Exact arithmetic (fractions, radicals) is expected where the numbers permit it.

Problem 1. Work on the uniform knots $\tau_i = i$, $i \in \mathbb{Z}$, with the order-2 B-splines (hat functions)

$$N_{i2}(x) = \begin{cases} x - i, & i \le x < i+1, \\ i + 2 - x, & i+1 \le x < i+2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Verify the partition of unity directly: for any $x \in [j+1,\, j+2)$, identify the (at most two) B-splines that are nonzero at x and show that their values sum to 1.

(b) For $S(x) = \sum_i \gamma_i N_{i2}(x)$, show that $S(\tau_{j+1}) = \gamma_j$: order-2 spline curves interpolate their own coefficients at the knots.

(c) Explain why (b) fails for cubic B-splines ($k = 4$): how many B-splines are nonzero at an interior knot, and what does S at a knot equal instead?

Problem 2. Consider quadrature rules on $[0, 1]$ with the three nodes $0, \frac{1}{2}, 1$:

$$\int_0^1 f(x)\,dx \;\approx\; w_0 f(0) + w_1 f(\tfrac{1}{2}) + w_2 f(1).$$

(a) Determine w_0, w_1, w_2 by requiring exactness for $f = 1,\, x,\, x^2$, and identify the resulting rule.

(b) Determine the exact degree of exactness of your rule by testing x^3 and x^4.

(c) The three nodes are fixed and symmetric. In one or two sentences, reconcile the answer of (b) with the parameter count: three weights, yet exactness beyond degree 2.

Problem 3. Construct the three-point Gaussian quadrature rule on $[-1, 1]$.

(a) Verify that $p_3(x) = x^3 - \frac{3}{5}x$ is orthogonal to $1, x, x^2$ in $L^2(-1, 1)$, and find its zeros (the Gaussian nodes).

(b) Determine the weights from exactness on 1 and x^2 (use the symmetry of the rule to reduce the work), and state the complete rule.

(c) Test the rule on $f(x) = x^6$: compute both the rule value and the exact integral, and explain why disagreement is consistent with the theory.

Problem 4. Let $x = (1, 2, 2)^T$.

(a) Construct the Householder reflector $Q = I - 2vv^T$ (with $\|v\|_2 = 1$) such that $Qx = \gamma e_1$, using the sign choice from lecture that avoids cancellation. Give γ, the unnormalized vector $\tilde{v}$, and the matrix Q with exact rational entries.

(b) Verify by direct multiplication that $Qx = \gamma e_1$.

(c) Verify that Q is symmetric and orthogonal, and determine its eigenvalues (with multiplicities) without computing a characteristic polynomial.

Problem 5. Let $\epsilon > 0$ be small and

$$A = \begin{bmatrix} 2 & \epsilon \\ \epsilon & 1 \end{bmatrix},$$

a symmetric matrix with eigenvalues $\lambda_1 = 2 + O(\epsilon^2)$ and $\lambda_2 = 1 + O(\epsilon^2)$.

(a) Carry out one step of the *unshifted* QR iteration ($A = QR$, $A' = RQ$) and show that the new off-diagonal entry satisfies $a'_{21} = \frac{\epsilon}{2} + O(\epsilon^3)$. Working to leading order in ϵ is acceptable throughout.

(b) Interpret the factor $\frac{1}{2}$ in terms of the convergence theory of the QR algorithm.

(c) Now carry out one step of the *shifted* iteration with $\mu = a_{22} = 1$ (factor $A - \mu I = QR$, set $A' = RQ + \mu I$), and show that the new off-diagonal entry is $O(\epsilon^3)$. What property of the shifted algorithm for symmetric matrices does this illustrate?